

NFC CARD READER

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PROJECT OVERVIEW

Giantec Semiconductor Corporation | Software Engineer Intern | May–June 2025

Developed cross-platform Android and iOS applications that use built-in mobile NFC capabilities to read and write NDEF-formatted data. The applications allow users to view stored tag information, encode and write custom fields, and review timestamped operation logs.

In addition to mobile application development, I worked with chip design engineers to understand and validate the underlying NFC communication process. After training, I independently reproduced an MCU-based hardware test workflow and performed RF waveform and I²C protocol verification.

Technologies: Java · Android Studio · Swift · SwiftUI · Core NFC · NDEF



Android interface shown. The corresponding iOS application implemented the same feature set and workflow using Swift and SwiftUI.

FEATURES

*Interface language and labels reflected the internal prototype used during development.

Connects to a nearby NFC tag using the green control and disconnects the active NFC connection using the red control. The detected tag technology is displayed above.



Records NFC operations in a scrollable log and displays the total number of recorded entries for easier testing and debugging.

Provides a simple navigation structure for accessing the main NFC workflow and application settings.

Supports writing custom data to NFC tags, reading stored data blocks, and clearing recorded operation logs.

DEVELOPMENT AND TESTING

CROSS-PLATFORM IMPLEMENTATION

Developed corresponding Android and iOS applications with the same core workflow and feature set. The Android version was implemented in Java, while the iOS version was developed using Swift and SwiftUI.

PHYSICAL-DEVICE TESTING

Tested the applications on physical Android and iOS devices to validate NFC tag detection, NDEF reading and writing, interface responses, and operation logging.

DEBUGGING AND VALIDATION

Used operation logs and status messages to identify NFC read/write issues, test error-handling behavior, and verify expected application responses.

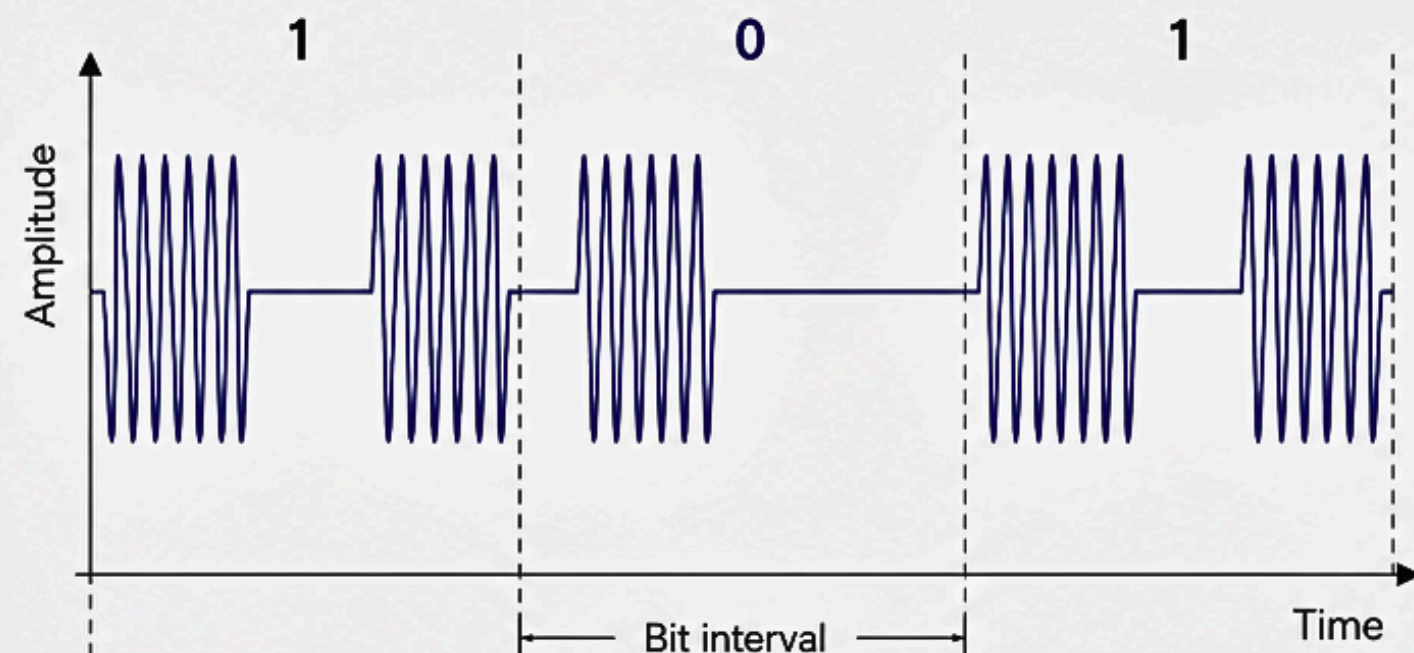
My Contributions

- Developed and tested Android and iOS NFC applications
- Implemented NDEF reading, writing, custom data encoding, and operation logging
- Designed mobile user interfaces and validated workflows on physical devices
- Verified RF and I²C communication using an MCU-based test platform, oscilloscope, and logic analyzer.

HARDWARE-LEVEL VALIDATION

RF Waveform Analysis (13.56 MHz)

Modified Miller Coding (Reader → Tag)



Modified Miller Rules



Pause in the middle
= 1



Pause at the beginning
= 0

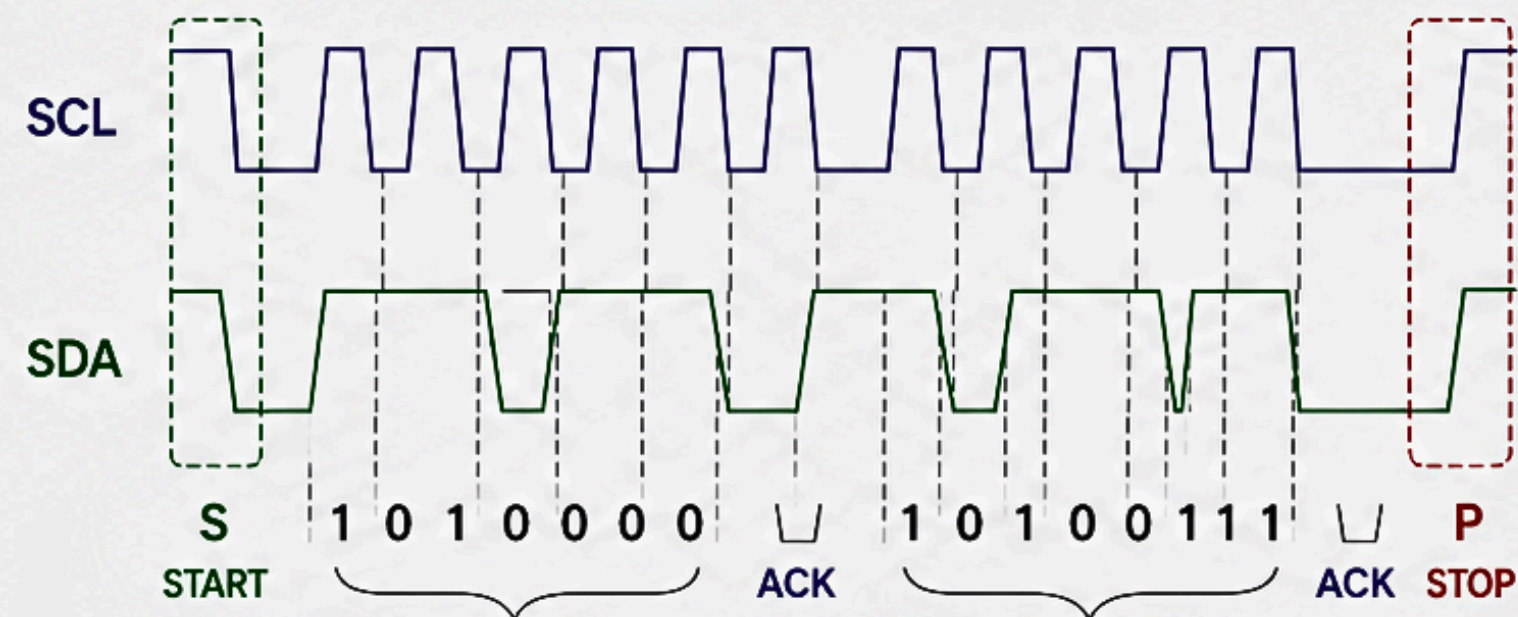


No pause
(depends on
previous bit)

Manually decoded RF bit values from carrier-pause timing within fixed bit intervals.

I²C Protocol Analysis

SDA (Data) and SCL (Clock)



ADDRESS → ACK → DATA → ACK

I²C Decode:

START	0x50 (W)	ACK	0xA7	ACK	STOP
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Inspected SDA/SCL timing, START/STOP conditions, data bytes, and ACK responses using a logic analyzer.